

**Ime in Priimek:** _____**Naloga 1**

Zapiši s kotnimi funkcijami ostrih kotov v kotnih stopinjah in izračunaj:

a)
$$\frac{\cos 405^\circ + \sin(-150^\circ)}{\tan(-765^\circ) \cdot \cot(-240^\circ)}$$

b)
$$\frac{\sin(1110^\circ - \cos 495^\circ)}{\tan(-675^\circ) + \cot^2 300^\circ}$$

c)
$$\frac{\cos(-300^\circ) \cdot \sin 480^\circ}{\tan 1140^\circ - \cot 135^\circ}$$

d)
$$\frac{\cos 1080^\circ - \sin(-570^\circ)}{\frac{1}{2} \tan 225^\circ + \cot(-390^\circ)} + \sin^2(-30^\circ)$$

e)
$$\frac{\sin^2 750^\circ + \cos 225^\circ}{\tan(-570^\circ) \cdot \cot 330^\circ}$$

Naloga 2

Zapiši s kotnimi funkcijami ostrih kotov v radianih in izračunaj:

$$\text{a) } \frac{\sin \frac{17\pi}{4} + \cos \left(-\frac{5\pi}{3}\right)}{\tan \left(-\frac{9\pi}{2}\right) \cdot \cot^2 \left(\frac{5\pi}{6}\right)}$$

$$\text{d) } \frac{\cos \left(-\frac{10\pi}{3}\right) + \sin \frac{23\pi}{4}}{\tan \frac{17\pi}{6} \cdot \cot \left(-\frac{5\pi}{4}\right)} + \sin \frac{13\pi}{6}$$

$$\text{b) } \frac{\cos \frac{25\pi}{6} - \sin \left(-\frac{7\pi}{4}\right)}{\tan \frac{19\pi}{3} + \cot \frac{3\pi}{4}} \cdot \left(\tan \frac{4\pi}{3} - \cot \frac{5\pi}{4}\right)$$

$$\text{e) } \frac{\sin \frac{29\pi}{6} - \cos \frac{19\pi}{4}}{\tan \left(-\frac{11\pi}{3}\right) + \cot^2 \frac{7\pi}{4}}$$

$$\text{c) } \frac{\sin^2 \frac{13\pi}{3} \cdot \cos \left(-\frac{11\pi}{6}\right)}{\tan \frac{15\pi}{4} - \cot \frac{4\pi}{3}}$$

Naloga 3

Zavrti točko $A\left(-\frac{1}{2}, -\frac{\sqrt{3}}{2}\right)$ za pravi kot okoli koordinatnega izhodišča v smeri urnega kazalca. Določi koordinate te točke.

Naloga 4

Na krožnici s polmerom 2 leži točka $T(\sqrt{3}, y < 0)$.

a) Kakšni koordinati ima ta točka, če a) jo zavrtimo okoli koordinatnega izhodišča v nasprotni smeri gibanja urnega kazalca za kot 165° ?

b) če bi jo zavrteli v smeri gibanja urnega kazalca za kot 450° ?

Naloga 5

Poenostavi:

a) $(1 + \tan^2 x)(1 - \sin^2 x)$

b) $(1 + \cot^2 x)(1 - \cos^2 x)$

c) $(3 \cos x + 2 \sin x)^2 + (3 \cos x - 2 \sin x)^2$

d) $\frac{1 + \cos x}{\sin x} - \frac{\sin x}{1 - \cos x}$

e) $\sin\left(x + \frac{\pi}{3}\right) + \sin\left(x - \frac{7\pi}{3}\right)$

f) $\cos\left(x + \frac{\pi}{4}\right) + \cos\left(x - \frac{9\pi}{4}\right)$

g) $\tan\left(x + \frac{\pi}{4}\right) \cdot \tan\left(x - \frac{5\pi}{4}\right)$

Naloga 6

Poenostavi:

a) $\frac{\sin 2x}{1 + \cos 2x}$

b) $\frac{\cot^2 x - 1}{2 \cot x} \cdot \frac{\sin 2x}{2 \cos^2 x - 1}$

c) $\cot x - \tan x - 2 \cot 2x$

Naloga 7

a) Naj velja za oster kot x : $\tan x = \frac{12}{5}$. Izračunaj $\sin 2x$ in $\cos(x - \frac{\pi}{4})$.

b) Naj bo $0 < x < \pi$ in $\cot x = \frac{3}{4}$. Izračunaj $\cos 2x$ in $\tan(x - \frac{\pi}{4})$.

c) Naj bo $\pi < x < 2\pi$ in $\cos x = \frac{\sqrt{5}}{3}$. Izračunaj $\tan 2x$ in $\cot(x + \frac{\pi}{4})$.

d) Naj bo x topi kot in $\sin x = \frac{\sqrt{15}}{4}$. Izračunaj $\sin(x + \frac{\pi}{4})$ in $\cos(x - \frac{5\pi}{4})$.

e) Naj za ostru kota x in y velja $\sin x = \frac{3}{5}$ in $\cos y = \frac{5}{13}$. Izračunaj $\tan(x - y)$, $\sin \frac{x}{2}$, $\tan \frac{y}{2}$.

f) Naj bo $\frac{3\pi}{2} < x < 2\pi$ in $\sin 2x = -\frac{24}{25}$. Izračunaj $\sin x$, $\sin \frac{x}{2}$.

g) Naj bo $\frac{\pi}{2} < x < \pi$ in $\cos 2x = \frac{40}{41}$. Izračunaj $\cos x$, $\cos \frac{x}{2}$.

h) Naj bo x oster kot in $\tan 2x = \frac{7}{24}$. Izračunaj $\tan x$, $\tan \frac{x}{2}$.

Naloga 8

Dokaži identiteto:

a) $\frac{\sin(3x)}{\sin x} - \frac{\cos(3x)}{\cos x} = 2$

b) $\sin^4 x = \frac{3 - 4 \cos(2x) + \cos(4x)}{8}$

c) $\tan(3x) = \frac{3 \tan x - \tan^3 x}{1 - 3 \tan^2 x}$

d) $\cos 4x = 8 \cos^4 x - 8 \cos^2 x + 1$

Naloga 9

Dokaži enakost:

a) $\frac{\cos x \sin 2x + \cos 2x + 1}{\sin x + 1} \cdot (1 + \tan^2 x) = 2$

b) $\frac{\cos x + \cos 2x + 1}{\sin x + \sin 2x} = \cot x$

c) $\frac{\sin 2x}{\sin x} (1 + \tan^2 x) = \frac{2}{\cos x}$

Naloga 10

Zapiši kot produkt in izračunaj:

a) $\cos 105^\circ - \cos 15^\circ$

d) $\sin 5^\circ + \sin 55^\circ - \cos 25^\circ$

b) $\sin 105^\circ + \sin 15^\circ$

e) $\sin 10^\circ + \sin 50^\circ - \cos 20^\circ$

c) $\cos 15^\circ - \sin 15^\circ$

f) $\sin 8^\circ + \cos 38^\circ + \cos 158^\circ$

Naloga 11

Zapiši kot vsoto ali razliko:

a) $\cos 7x \cos 5x$

d) $\sin(2x + 3y) \sin(2x - 3y)$

b) $\sin 2x \sin 7x$

e) $\cos(x + y) \cos(x - y)$

c) $\sin 6x \cos 4x$

f) $2 \sin(4x + y) \cos(4x - y)$

Naloga 12

Pokaži, da v trikotniku velja: $\sin 2\alpha + \sin 2\beta + \sin 2\gamma = 4 \sin \alpha \sin \beta \sin \gamma$

Naloga 13

Dokaži enakost:

a) $\frac{2 \sin 5x \cdot \sin 4x}{\cos x - \cos 9x} = 1$

b) $\frac{\cos 2x \cos 10x}{\cos 12x + \cos 8x} = \frac{1}{2}$

c) $\frac{\sin 4x + \sin 8x}{\cos 4x - \cos 8x} = \cot 2x$

Naloga 14

Nariši grafe funkcij:

a) $f(x) = 2 \cos(x + \frac{\pi}{4}) - 1$

b) $f(x) = -2 \sin(x + \pi) - 1$

c) $f(x) = \tan\left(\frac{x}{2} - \frac{\pi}{4}\right)$

a) $f(x) = |\cos(x - \frac{\pi}{3})| - 1$

b) $f(x) = -|\sin\left(\frac{x}{2} + \frac{\pi}{4}\right) + 2|$

c) $f(x) = |\cot\left(\frac{x}{3} - \frac{\pi}{6}\right)|$

Naloga 15

Reši enačbo:

a) $\cos(x + \frac{\pi}{2}) = \frac{\sqrt{2}}{2}$

b) $2 \sin(x + \pi) = -\sqrt{3}$

c) $-3 \tan(\frac{x}{3} + \frac{\pi}{6}) = \sqrt{3}$

d) $\cot(\frac{x}{2} + \frac{\pi}{3}) = \sqrt{3}$

e) $\cot^2 x - \frac{1}{9} = 0$

f) $\sin^2 x - \cos x = 1$

g) $\sin^2 x - \sin x = 2$

h) $\sin x = \sqrt{3} \cos x$

i) $\sin 2x \tan x = \frac{3}{2}$

j) $\cos 3x \sin 2x - \cos 3x = 0$

k) $\sin^2 x + 2 \sin x \cos x - 3 \cos^2 x = 1$

l) $\sin^2 x - 6 \sin x \cos x - \cos^2 x = 3$

m) $\cos 2x = 5 \sin^2 x - \cos^2 x$

n) $\cos^3 x - \cos x = \sin^2 x$

$$\begin{aligned}
1. \text{ Kote prevedemo v osnovne: } -405^\circ &= 360^\circ + 45^\circ \Rightarrow \cos(405^\circ) = \cos(45^\circ) = \frac{\sqrt{2}}{2} - \sin(-150^\circ) = \\
&= -\sin(150^\circ) = -\sin(180^\circ - 30^\circ) = -\sin(30^\circ) = -\frac{1}{2} - 765^\circ = 2 \cdot 360^\circ + 45^\circ \Rightarrow \tan(765^\circ) = \tan(45^\circ) = 1 - \\
240^\circ &= 180^\circ + 60^\circ \Rightarrow \cot(240^\circ) = \cot(60^\circ) = \frac{\sqrt{3}}{3}
\end{aligned}$$

$$\text{Števec: } \frac{\sqrt{2}}{2} + \left(-\frac{1}{2}\right) = \frac{\sqrt{2}-1}{2}$$

$$\text{Imenovalec: } 1 \cdot \frac{\sqrt{3}}{3} = \frac{\sqrt{3}}{3}$$

Celoten izraz:

$$\frac{\frac{\sqrt{2}-1}{2}}{\frac{\sqrt{3}}{3}} = \frac{\sqrt{2}-1}{2} \cdot \frac{3}{\sqrt{3}} = \frac{3(\sqrt{2}-1)}{2\sqrt{3}} = \frac{\sqrt{3}(\sqrt{2}-1)}{2}$$

$$\boxed{\frac{\sqrt{3}(\sqrt{2}-1)}{2}}$$

$$\begin{aligned}
2. \text{ Kote prevedemo: } -1110^\circ &= 3 \cdot 360^\circ + 30^\circ \Rightarrow \sin(1110^\circ) = \sin(30^\circ) = \frac{1}{2} - 495^\circ = 360^\circ + 135^\circ \Rightarrow \cos(495^\circ) = \\
\cos(135^\circ) &= -\frac{\sqrt{2}}{2} - 675^\circ = -2 \cdot 360^\circ + 45^\circ \Rightarrow \tan(-675^\circ) = \tan(45^\circ) = 1 - 300^\circ = 360^\circ - 60^\circ \Rightarrow \\
\cot(300^\circ) &= -\cot(60^\circ) = -\frac{\sqrt{3}}{3} \Rightarrow \cot^2(300^\circ) = \frac{1}{3}
\end{aligned}$$

$$\text{Števec: } \frac{1}{2} - \left(-\frac{\sqrt{2}}{2}\right) = \frac{1+\sqrt{2}}{2}$$

$$\text{Imenovalec: } 1 + \frac{1}{3} = \frac{4}{3}$$

Celoten izraz:

$$\frac{\frac{1+\sqrt{2}}{2}}{\frac{4}{3}} = \frac{1+\sqrt{2}}{2} \cdot \frac{3}{4} = \frac{3(1+\sqrt{2})}{8}$$

$$\boxed{\frac{3(1+\sqrt{2})}{8}}$$

$$\begin{aligned}
3. \text{ Kote prevedemo: } -\cos(-300^\circ) &= \cos(300^\circ) = \cos(360^\circ - 60^\circ) = \cos(60^\circ) = \frac{1}{2} - 480^\circ = 360^\circ + 120^\circ \Rightarrow \\
\sin(480^\circ) &= \sin(120^\circ) = \sin(60^\circ) = \frac{\sqrt{3}}{2} - 1140^\circ = 3 \cdot 360^\circ + 60^\circ \Rightarrow \tan(1140^\circ) = \tan(60^\circ) = \sqrt{3} - \\
135^\circ &= 180^\circ - 45^\circ \Rightarrow \cot(135^\circ) = -\cot(45^\circ) = -1
\end{aligned}$$

$$\text{Števec: } \frac{1}{2} \cdot \frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{4}$$

$$\text{Imenovalec: } \sqrt{3} - (-1) = \sqrt{3} + 1$$

Celoten izraz:

$$\frac{\frac{\sqrt{3}}{4}}{\sqrt{3}+1} = \frac{\sqrt{3}}{4(\sqrt{3}+1)} \cdot \frac{\sqrt{3}-1}{\sqrt{3}-1} = \frac{3-\sqrt{3}}{4(3-1)} = \frac{3-\sqrt{3}}{8}$$

$$\boxed{\frac{3-\sqrt{3}}{8}}$$

$$\begin{aligned}
4. \text{ Kote prevedemo: } -750^\circ &= 2 \cdot 360^\circ + 30^\circ \Rightarrow \sin(750^\circ) = \sin(30^\circ) = \frac{1}{2} \Rightarrow \sin^2(750^\circ) = \frac{1}{4} - 225^\circ = \\
180^\circ + 45^\circ &\Rightarrow \cos(225^\circ) = -\cos(45^\circ) = -\frac{\sqrt{2}}{2} - 540^\circ = -2 \cdot 360^\circ + 180^\circ \Rightarrow \tan(-540^\circ) = \tan(180^\circ) = 0 \\
-330^\circ &= 360^\circ - 30^\circ \Rightarrow \cot(330^\circ) = -\cot(30^\circ) = -\sqrt{3}
\end{aligned}$$

$$\text{Števec: } \frac{1}{4} + \left(-\frac{\sqrt{2}}{2}\right) = \frac{1-2\sqrt{2}}{4}$$

Imenovalec: $0 \cdot (-\sqrt{3}) = 0$

Deljenje z nič je nedefinirano.

Nedefinirano

5. Kote prevedemo: $-1080^\circ = 3 \cdot 360^\circ + 0^\circ \Rightarrow \cos(1080^\circ) = \cos(0^\circ) = 1$ - $\sin(-570^\circ) = -\sin(570^\circ) = -\sin(360^\circ + 210^\circ) = -\sin(210^\circ) = -(-\sin(30^\circ)) = \frac{1}{2}$ - $150^\circ = 180^\circ - 30^\circ \Rightarrow \tan(150^\circ) = -\tan(30^\circ) = -\frac{\sqrt{3}}{3}$ - $420^\circ = 360^\circ + 60^\circ \Rightarrow \cot(420^\circ) = \cot(60^\circ) = \frac{\sqrt{3}}{3} \Rightarrow \cot^2(420^\circ) = \frac{1}{3}$

Števec: $1 - \frac{1}{2} = \frac{1}{2}$

Imenovalec: $-\frac{\sqrt{3}}{3} + \frac{1}{3} = \frac{1 - \sqrt{3}}{3}$

Celoten izraz:

$$\frac{\frac{1}{2}}{\frac{1-\sqrt{3}}{3}} = \frac{1}{2} \cdot \frac{3}{1-\sqrt{3}} = \frac{3}{2(1-\sqrt{3})} \cdot \frac{1+\sqrt{3}}{1+\sqrt{3}} = \frac{3(1+\sqrt{3})}{2(1-3)} = \frac{3(1+\sqrt{3})}{-4} = -\frac{3(1+\sqrt{3})}{4}$$

$$-\frac{3(1+\sqrt{3})}{4}$$